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19 June 1950
SUPERSEDING
Picatinny Arsenal
PXS-1289
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MILITARY SPECIFICATION

CUTTER, REEFING LINE, M2

1. SCOPE

1.1 Scope.--This specification covers the following reefing line cutters for use with cargo parachutes released from high speed aircraft:

Two seconds delay.
Four seconds delay.
Six seconds delay.
Eight seconds delay.
Ten seconds delay.

2. APPLICABLE SPECIFICATIONS AND DRAWINGS

2.1 Specifications.--All specifications referenced on drawings listed in 2.2, of the issue in effect on date of invitation for bids, form a part of this specification.

(Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Drawings.--The following drawing and all drawings referenced thereon, of the issue in effect on date of invitation for bids, form a part of this specification:

ORDNANCE DEPARTMENT DRAWING

82-O-159 - Cutter, Reefing Line, M2, Assembly.

(Copies of drawings should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Material.--Materials and parts shall be in accordance with applicable drawings and specifications.

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3.2 The cutters shall meet all requirements specified on the drawings and in applicable specifications.

3.3 Black powder.—The moisture content of the black powder at the time of loading shall not exceed the amount prescribed in Specification JAN-P-223.

3.4 Non-gaseous powder.—

3.4.1 Delay (applicable to four, six, eight, and ten seconds delay only).—The moisture content of the non-gaseous powder (delay) at the time of loading shall not exceed .20 percent.

3.4.2 Igniter.—The moisture content of the non-gaseous powder (igniter) at the time of loading shall not exceed .20 percent (see 4.5).

3.5 Hydrostatic.—No knife-delay element shall show any evidence of leakage when tested as prescribed in 4.6.

3.6 Functioning.—No cutter shall fail to fire within the prescribed pull requirements or fail to fire within the delay limits prescribed on the drawing, or fail to cut the reefing line, when tested as prescribed in 4.7.

3.7 High temperature.—No cutter shall fail to fire and cut the reefing line when tested at a temperature of 160° F. (see 4.8).

3.8 Low temperature.—No cutter shall fail to fire and cut the reefing line when tested at a temperature of -65° F. (see 4.9).

3.9 Workmanship.—All parts shall be free of chips, dirt, grease, and other foreign material. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents used.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.—A lot shall consist of 10,000 cutters, maximum. A smaller quantity may constitute a lot at the discretion of the inspector. Two thousand plus or minus 500 shall constitute the first lot manufactured on any contract, except when a new contract immediately follows the manufacture of satisfactory material of the same character. Each lot shall contain cutters from one class only and primers from not more than one lot.

4.2 Sampling.—All samples shall be selected by the inspector. Forty cutters shall be selected from each lot for tests as specified in the following paragraphs. Sampling of component materials and components is prescribed in paragraphs covering tests of the materials and components.

4.3 Inspection.—Inspection shall be as specified in Specification 50-0-1.

4.4 Determination of moisture content of black powder.—The moisture content of the black powder shall be determined on each lot or sub-lot at the time of loading. The allowable moisture content and the method of determining the percentage thereof shall be in accordance with Specification JAN-P-223.

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4.5 Test of non-gaseous powder.-The moisture content of the non-gaseous powder (delay or igniter) shall be determined on each lot or sub-lot at the time of loading. The method of determination shall be as follows:

4.5.1 Moisture content.-Transfer an accurately weighed portion of approximately 1 gram of delay (or igniter) powder to a tared 50-ml beaker containing a medium-porosity sintered-glass filter stick. Place beaker and contents in a drying oven at 100° to 105°C. for 2 hours. Cool in a desiccator and weigh. Calculate the percentage of moisture in the sample as follows:

$$\text{Percentage of moisture} = \frac{100 \times A}{W}$$

where:

A = loss in weight of beaker and contents.
W = weight of sample.

4.5.2 Composition.-

4.5.2.1 Potassium perchlorate.-To the dried sample from the moisture determination in the 50 ml beaker, add 20 ml of hot water. Stir the mixture for approximately 5 minutes and then apply suction to the filter stick until most of the liquid in the beaker has drained off. Repeat with one 20-ml portion and then with two 10-ml portions of hot water. Place beaker and contents in a drying oven at 100° to 105°C. for two hours. Cool in a desiccator and weigh. Calculate the loss in weight of the beaker and contents to percentage of potassium perchlorate in the sample as follows:

$$\text{Percentage of potassium perchlorate} = \frac{100 \times A}{W}$$

where:

A = loss in weight of beaker and contents.
W = weight of sample.

4.5.2.2 Nickel.-To the residue from the potassium perchlorate extraction in the 50-ml beaker, add 20 ml of 8 N hydrochloric acid. Boil gently for one half hour. Dilute to approximately 40 ml. With the aid of suction remove the solution by means of the filter stick. Wash the residue with three 20-ml portions of water. Reserve the residue for the determination of zirconium and graphite. In a 250-ml volumetric flask dilute the filtrate and washings to the mark. Take a 25-ml aliquot and dilute to approximately 250 ml in a 400-ml beaker. Add 3 grams of tartaric acid and 10 ml of a 10-percent ammonium chloride solution. Add 15 N ammonium hydroxide until the solution is slightly alkaline. If a precipitate forms, add more ammonium chloride to clear the solution and again make just alkaline with 15 N ammonium hydroxide.

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Repeat this process until the solution is clear when slightly alkaline. Heat nearly to boiling and add 10 ml of 1-percent solution of dimethylglyoxime in ethanol. Make the solution distinctly ammoniacal and stir. Place beaker and contents on a steam bath for 15 to 20 minutes stirring frequently. Filter, using a tared Gooch crucible. After the precipitate has been transferred to the crucible, wash it with two 20-ml portions of hot water. Dry the precipitate for 2 hours at 110° to 120°C. Cool in a desiccator and weigh. Calculate the percentage of nickel in the sample as follows:

$$\text{Percentage of nickel} = \frac{203.2 \times A}{W}$$

where:

A = gain in weight of crucible.
W = weight of sample.

4.5.2.3 Barium chromate.-Place the beaker containing the filter stick and the residue from the extraction with 8 N hydrochloric acid in a drying oven at 100° to 105°C. for 2 hours. Cool in a desiccator and weigh. Calculate the loss in weight of the beaker and contents to percentage of barium chromate in the sample as follows:

$$\text{Percentage of barium chromate} = \frac{100 \times A}{W} - B$$

where: A = loss in weight of crucible and contents resulting from the hydrochloric acid treatment.
W = original weight of sample.
B = percentage of nickel.

4.5.2.4 Zirconium.-To the beaker and residue from the determination of barium chromate add 10 ml of 36 N sulfuric acid and 3 grams of potassium sulfate. Evaporate to fumes of sulfur trioxide and continue to fume for 15 minutes. Cool and add 20 ml of water. With the aid of suction and filtering stick remove the solution. Wash the residue with 10 ml of water and again add 5 ml of 36 N sulfuric acid and 1.5 grams of potassium sulfate and evaporate to fumes of sulfur trioxide. Cool the mixture add 20 ml of water and remove the solution by means of the filter stick. Wash the residue with five or six 10-ml portions of water. Place the beaker and graphite in a drying oven at 100° to 105°C. for two hours. Cool in a desiccator and weigh. Repeat the sulfuric acid and potassium sulfate treatment until there is no further loss in weight of the residue. Determine the loss in weight of the beaker and contents and calculate the percentage of zirconium in the sample as follows:

$$\text{Percentage of zirconium} = \frac{100 \times A}{W}$$

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where:

A = loss in weight of beaker and contents.
W = weight of sample.

4.5.2.5 Graphite.-Subtract from the weight of beaker and graphite the weight of the beaker plus filtering stick and calculate the percentage of graphite in the sample as follows:

$$\text{Percentage of graphite} = \frac{100 \times A}{W}$$

where:

A = weight of graphite.
W = weight of sample.

4.6. Hydrostatic test.- Twenty sample knife-delay elements selected by the inspector from each shift's production, shall be tested by placing each sample element under water in a vacuum chamber; the water shall contain the wetting agent Nacconol NR (sodium alkyl aryl sulfonate), or equal. Evacuate the chamber to a minimum pressure differential of 5 psi. If any sample shows a continuous stream of air bubbles, the entire day's production represented by the sample shall be submitted to test. Any element which fails when the entire day's production is being tested shall be rejected. After being submitted to the hydrostatic test, the elements shall be rinsed thoroughly in fresh running water to remove the wetting agent, and then dried at room temperature.

4.7 Pull and functioning test.-Twenty sample cutters selected by the inspector from each lot shall be tested as follows:

4.7.1 Assemble the sample cutters in the test fixture shown on Figure 1. Pass the reefing line (Specification AN-W-10, Webbing; Tubular, Nylon, 1000-lb. tensile strength, 1/2-inch width) through the hole in the cutter. Suspend a 25-pound weight from each end of the reefing line. Adjust switches as follows: By means of the firing wire hold Switch 2 in open position. To close switch and start the timing clock, remove the firing wire. (see 4.7.2). By means of tension in the reefing line hold Switch 1 in closed position. Switch is opened and clock is stopped when reefing line is cut.

4.7.2 Apply in turn (at "W" on Fig. 1) a 10-pound weight and a 25-pound weight to the end of a line fastened to the firing wire. The weights shall be applied gradually to avoid jerking the line. Each weight shall be applied for a maximum 3.0 seconds.

4.7.3 Retest.-A retest, using double the number of samples used in the original test, may be made if requested by the contractor provided:

a. Not more than one cutter fired when a load of 10 pounds was applied. Any failures occuring during the retest shall be cause for rejection of the entire lot of cutters, subject to correction of the firing wire and resubmission for test.

b. Not more than one cutter failed to fire because of failure of the arming wire to pull out of the cutter when a load of 25 pounds was applied.

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Any failures occuring during the retest, shall be cause for rejection of the entire lot of cutters, subject to correction of the firing wire and resubmission for test.

c. Not more than one cutter failed to function within the delay time tolerances specified on the drawing. If any failures occur in the retest, the lot shall be rejected. A lot of rejected cutters may be resubmitted for test provided a new knife-delay element from an accepted day's production is assembled in each cutter of the rejected lot.

d. Not more than one cutter failed to cut the reefing line. If any failures occur in the retest, the lot shall be rejected. A lot of rejected cutters may be resubmitted for test provided a new knife-delay element from an accepted day's production is assembled in each cutter of the rejected lot.

4.8 High temperature test.-Ten sample cutters, selected by the inspector from each lot, shall be conditioned and tested as follows:

4.8.1 Procedure.-Subject the sample cutters to a temperature of 160° ± 2°F. for 48 hours. If the storage chamber is not sufficiently near the test apparatus to permit the cutters to be tested immediately, a vacuum thermos, at a temperature of 160° ± 2°F., may be used to transport the cutters to the test apparatus. In such a case the cutters shall be transferred to the vacuum thermos immediately upon removal from the storage chamber and shall be transported to the test apparatus and tested as quickly as practicable, as prescribed in paragraphs 4.7.1 and 4.7.2.

4.8.2 Acceptance.-The lot shall be considered acceptable provided that each cutter fired and cut the reefing line. If 9 of the 10 cutters meet this requirement, the lot shall be considered eligible for retest under the provisions of paragraph 4.8.3. If less than 9 of the 10 cutters meet this requirement, the lot shall be rejected.

4.8.3 Retest.- A retest, as provided for in paragraph 4.8.2, may be made if requested by the contractor, using 20 cutters. The lot shall be considered acceptable on retest provided the 20 cutters fire and cut the reefing line.

4.9 Low temperature test.- Ten sample cutters, selected by the inspector from each lot, shall be conditioned and tested as follows:

4.9.1 Procedure.-Subject the sample cutters to a temperature of -65° ± 2° F. for 48 hours. If the storage chamber is not sufficiently near the test apparatus to permit the cutters to be tested immediately, a vacuum thermos, at a temperature of -65° ± 2°F. may be used to transport the cutters to the test apparatus. In such a case, the cutters shall be transferred to the vacuum thermos immediately upon removal from the storage chamber and shall be transported to the test apparatus and tested as quickly as practicable, as prescribed in paragraphs 4.7.1 and 4.7.2.

4.9.2 Acceptance.-The lot shall be considered acceptable provided that each cutter fired and cut the reefing line. If 9 of the 10 cutters meet this requirement, the lot shall be considered eligible for retest under the provisions of paragraph 4.9.3. If less than 9 of the 10 cutters meet this requirement, the lot shall be rejected.

4.9.3 Retest.-A retest, as provided for in paragraph 4.9.2, may be made if requested by the contractor, using 20 cutters. The lot shall be considered acceptable on retest provided the 20 cutters fire and cut the reefing line.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be as specified on the applicable drawing.

5.2 Packing.-Packing shall be as specified on the applicable drawing.

5.3 Labeling and marking.- Labeling and marking shall be as specified on the applicable drawing.

5.4 Data cards.- Data cards, 5 x 8 inches, made of commercial manila tagboard, approximately 200 pounds ream weight, shall be prepared for each lot of accepted cutters to provide the information specified below. Quantities (approximately 50 cards per lot) and distribution of data cards shall be in accordance with instructions issued by the Office, Chief of Ordnance.

AMMUNITION DATA CARD

Kind: _____ Amm. Lot No. _____
AIC Symbol _____ Quantity in Lot _____
Drg. No.* _____ Spec. No.* _____
Manufactured by _____ Date _____
Contract No. _____ Order No. _____
Packed _____ in a box.
Remarks: _____

Certified to by _____Inspector.
Print here name of inspector, district, arsenal, loading plant, and subdistrict where applicable.

*Include revision and amendment.

6. NOTES

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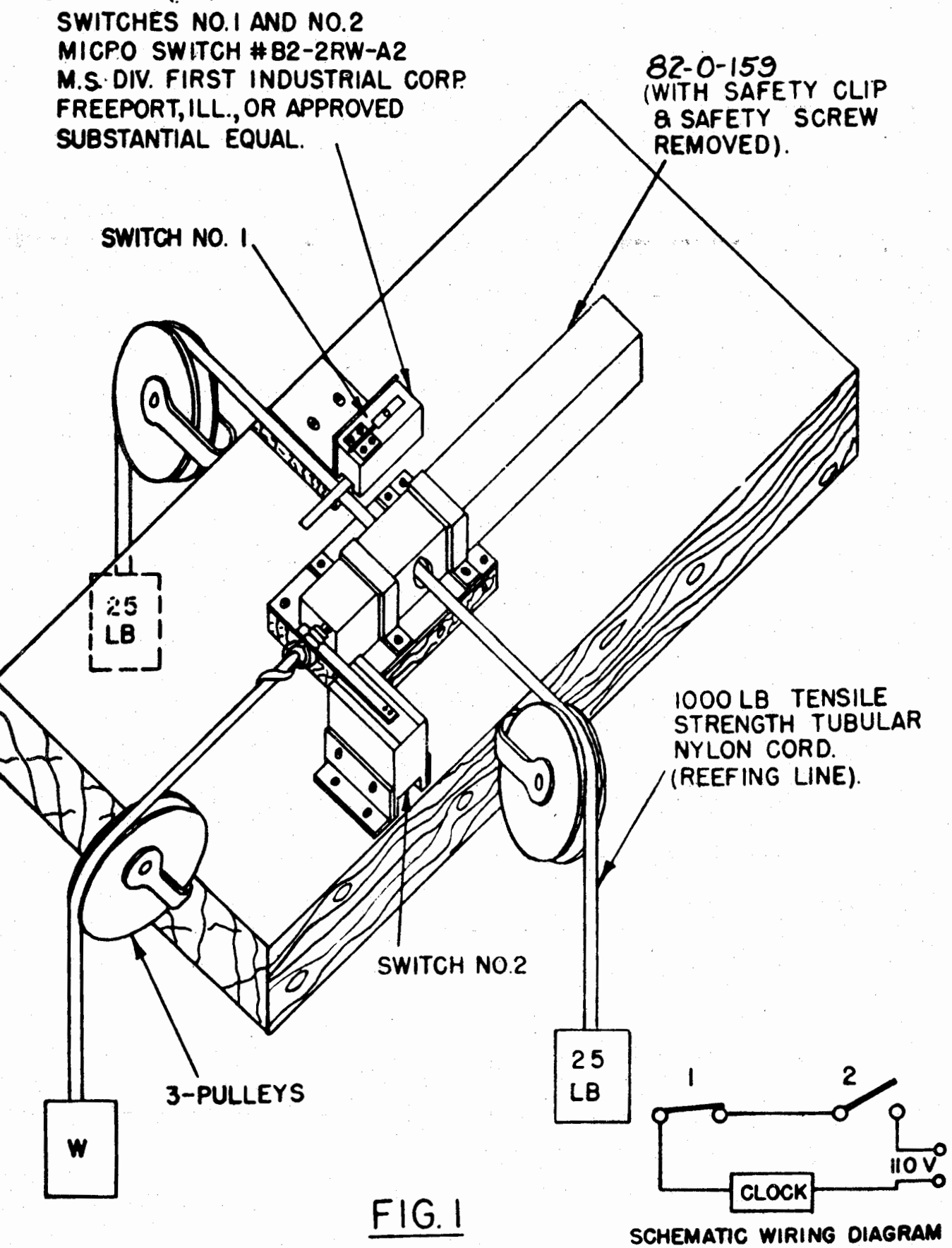


FIG. 1

SCHEMATIC WIRING DIAGRAM

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